Variational autoencoders

Lecture 6

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Which faces are fake?

Take a moment to look. Choose all that apply.

A



B



C



D



Which details influenced your choice?

Image credits appear with the answer on the next slide.

The answer

A



Photograph

B



Generated

C



Generated

D



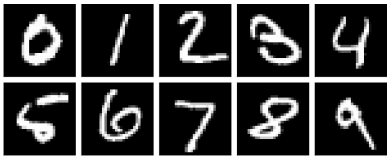
Photograph

B and C were generated by FLUX 3 Image (Black Forest Labs, 2026).
A flow-matching model, closely related to diffusion models.

A, D: NVIDIA FFHQ teaser, Karras et al. (2019). B, C: Black Forest Labs, FLUX 3 Image gallery; architecture: FLUX 3 announcement, July 23, 2026. Portrait crops.

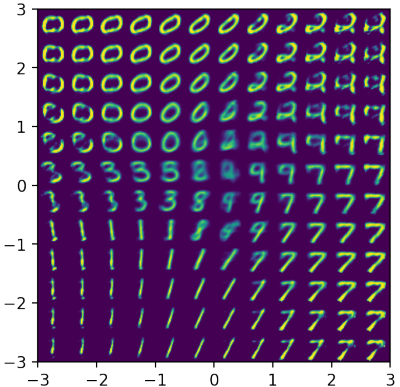
Are digits high-dimensional?

Real MNIST images



Each image is a vector of
$28 \times 28 = 784$ pixel values.

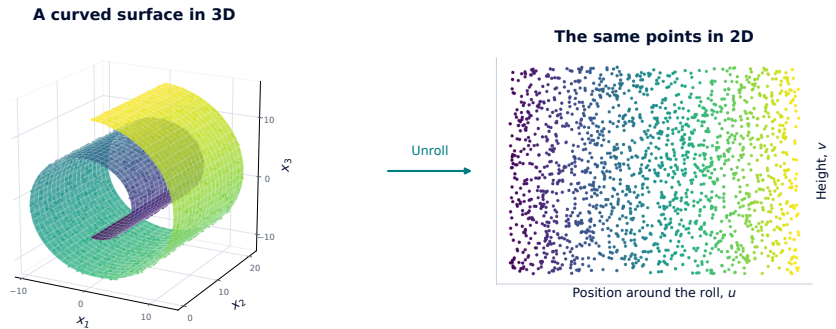
VAE-generated images



Images generated by a two-dimensional latent grid.

The manifold hypothesis

High-dimensional data often lie near a lower-dimensional surface.

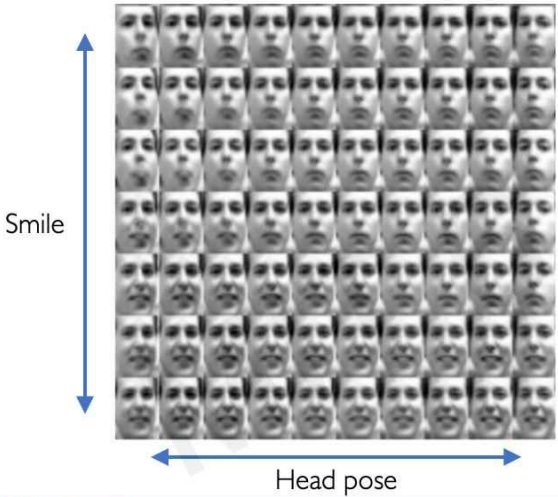


Ambient dimension: $D = 3$.
Each point has coordinates (x_1, x_2, x_3).

Intrinsic dimension: $d = 2$.
Position on the sheet needs only (u, v).

Swiss-roll example from scikit-learn (redrawn). Right: true underlying coordinates, not a fitted AE.

Few factors, many pixels



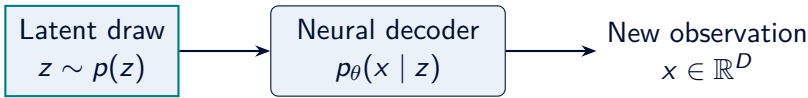
- Changing **head pose** or **expression** changes many pixels together.
- The image has many pixel coordinates, but some variation can be described by **a small number of factors**.
- Other factors include identity, lighting, and background.

Face-grid panel from Ava Amini, MIT 6.S191: Deep Generative Modeling (2026), slide 40. A latent-traversal illustration.

The generative idea

- $x \in \mathbb{R}^D$ is the high-dimensional observation (e.g., an image).
- $z \in \mathbb{R}^d$ is the low-dimensional latent variable ($d \ll D$).

Generative model:



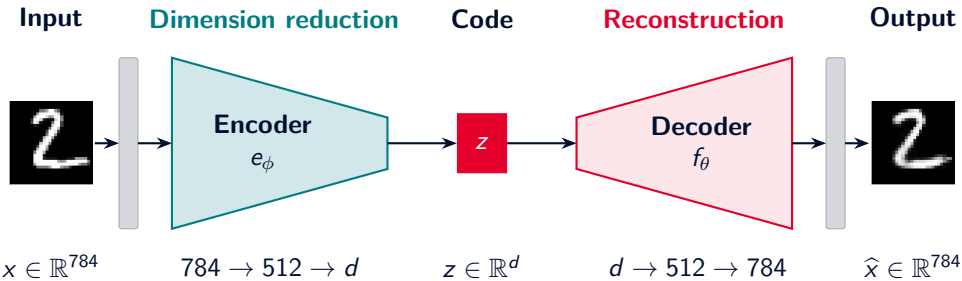
1. **Representation:** A low-dimensional latent variable can capture key factors of variation in high-dimensional data.
2. **Generation:** Draw a latent variable from a simple distribution, then use the neural decoder to generate a new observation.

How to learn a low-dimensional representation z from an observation x?

Notation follows the course VI lecture. VAE formulation: Kingma & Welling (2014).

Autoencoder architecture

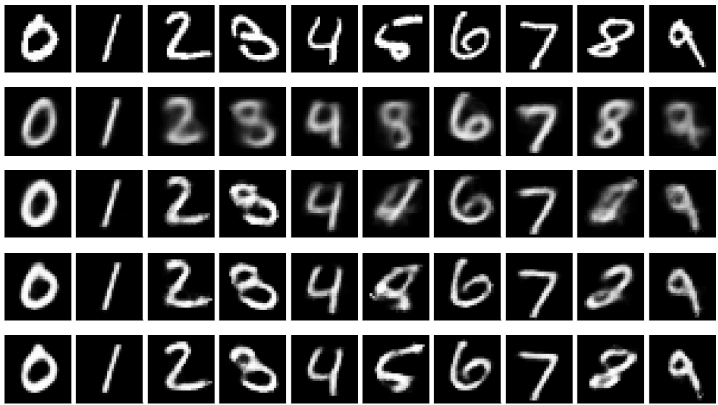
An **encoder** learns a compact code. A **decoder** reconstructs the input.



Our goal is:

$$\min_{\theta,\phi} \frac{1}{n} \sum_{i=1}^{n} \|x_i - \widehat{x}_i\|^2 \quad \text{where} \quad \widehat{x}_i = f_\theta(e_\phi(x_i))$$

The bottleneck dimension



Same input size

$D = 784.$

Different codes

$d = 2, 4, 8, 16.$

A wider bottleneck preserves more detail in these examples.

PCA as a linear autoencoder

Assume the data are centered; otherwise, first replace x_i by $x_i - \bar{x}$. Let $W \in \mathbb{R}^{D \times d}$ satisfy $W^\top W = I_d$.

$$\underbrace{z_i = W^\top x_i}_{\text{linear encoder}}, \qquad \underbrace{\widehat{x}_i = W z_i}_{\text{linear decoder}} .$$

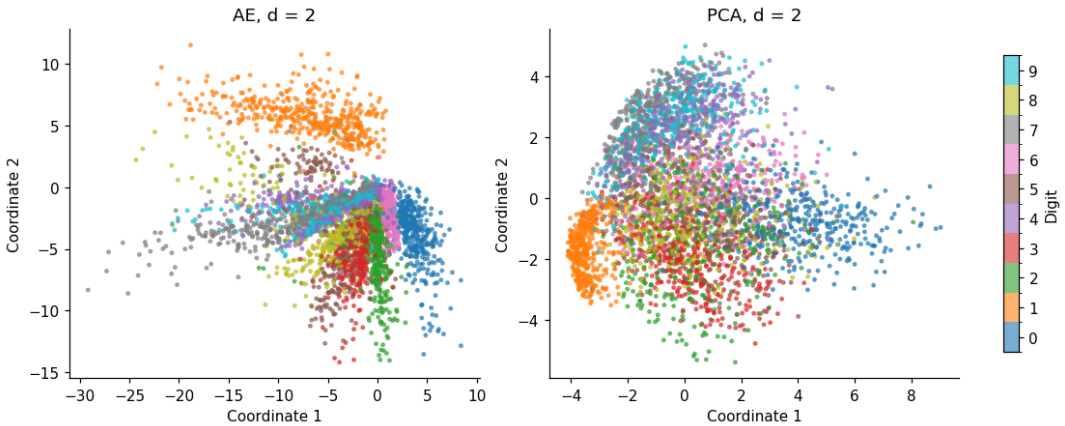
Minimizing reconstruction error is equivalent to maximizing retained variance:

$$\min_{W^\top W = I_d} \sum_{i=1}^{n} \|x_i - WW^\top x_i\|^2 \quad \Longleftrightarrow \quad \max_{W^\top W = I_d} \operatorname{tr}(W^\top S W), \quad \text{where} \quad S = \frac{1}{n} \sum_{i=1}^{n} x_i x_i^\top .$$

Solution: The columns of W are the top d eigenvectors of S.

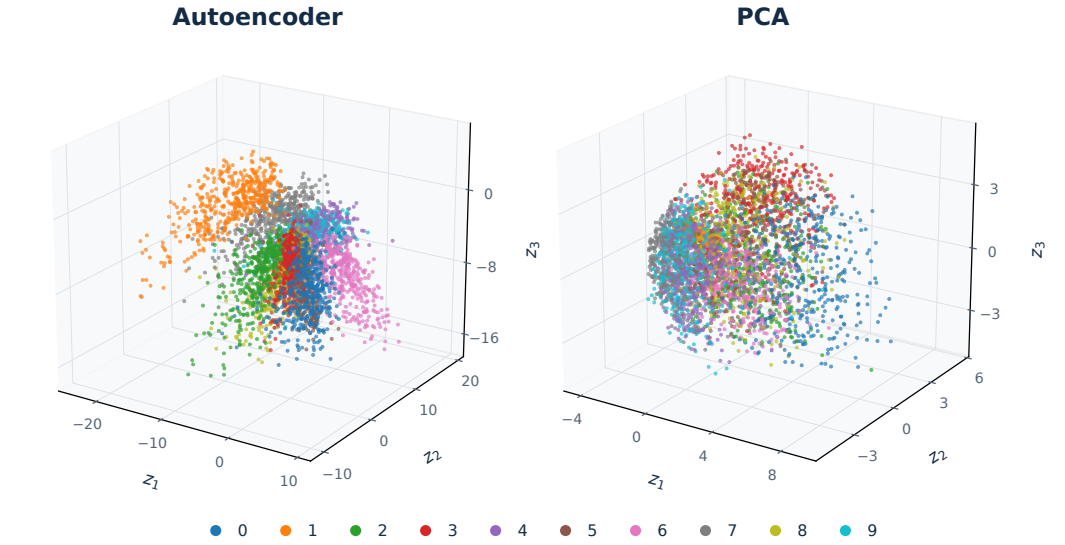
- **PCA:** uses linear mappings to encode and reconstruct data.
- **Autoencoder:** learns nonlinear encoder and decoder mappings, allowing more flexible representations.

Two-dimensional codes

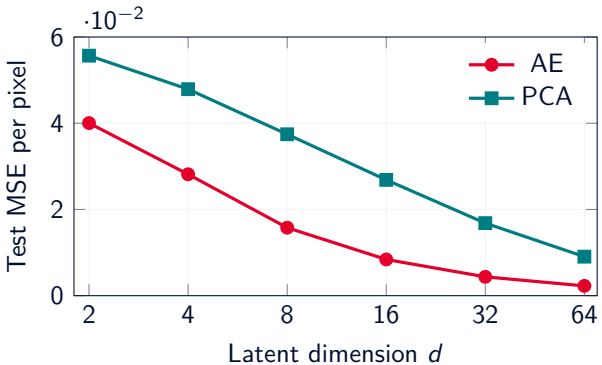


- Each dot is a test image.
- Color shows its digit label

Three-dimensional codes



Reconstruction error

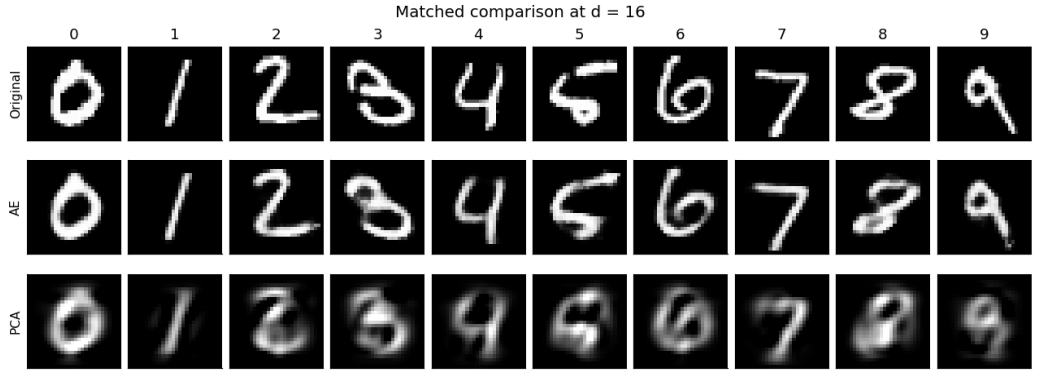


In this experiment:

- Both methods improve as d increases.
- The autoencoder (AE) has lower test MSE at each tested d.

Course MNIST AE/PCA notebook: 55,000 training images, 5,000 validation images, 10,000 test images. AE: 50 epochs per dimension. Values transcribed from the saved results table.

AE and PCA at $d = 16$



Both methods use 16 coordinates. The AE gives sharper reconstructions in these examples.

An ordinary AE is deterministic

At inference, an ordinary autoencoder applies two deterministic mappings:

$$x \xrightarrow{e_\phi} z \xrightarrow{f_\theta} \widehat{x}.$$

- **A single code:** For fixed network weights, each input x maps to one latent representation $z = e_\phi(x)$.
- **No sampling rule:** Reconstruction training alone does not specify a distribution $p(z)$ for generating new latent codes.
- **Toward probabilistic inference:** Specify a prior $p(z)$ and a decoder distribution $p_\theta(x \mid z)$; together, they define the posterior $p_\theta(z \mid x)$.

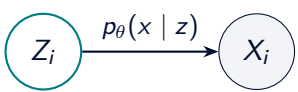
The **variational autoencoder (VAE)** connects latent inference with data generation.

The variational decoder

For observation i, introduce a latent variable $Z_i \in \mathbb{R}^d$:

$$Z_i \sim \mathcal{N}(0, I_d), \qquad X_i \mid Z_i = z_i \sim \mathcal{N}\Big(f_\theta(z_i), \tau^2 I_D\Big).$$

Here $f_\theta : \mathbb{R}^d \to \mathbb{R}^D$ is a neural decoder and $\tau^2 > 0$ is fixed.



Model sampling: Draw $z_{\text{new}} \sim \mathcal{N}(0, I_d)$, then

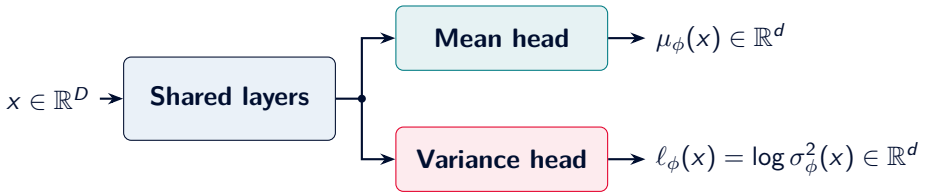
$$x_{\text{new}} = f_\theta(z_{\text{new}}) + \tau\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I_D).$$

For visualization in practice: Often display the decoder mean $f_\theta(z_{\text{new}})$ and omit the independent pixel noise $\tau\epsilon$.

The variational encoder

The nonlinear decoder usually gives an intractable posterior $p_\theta(z \mid x)$. Use the variational approximation as a Gaussian

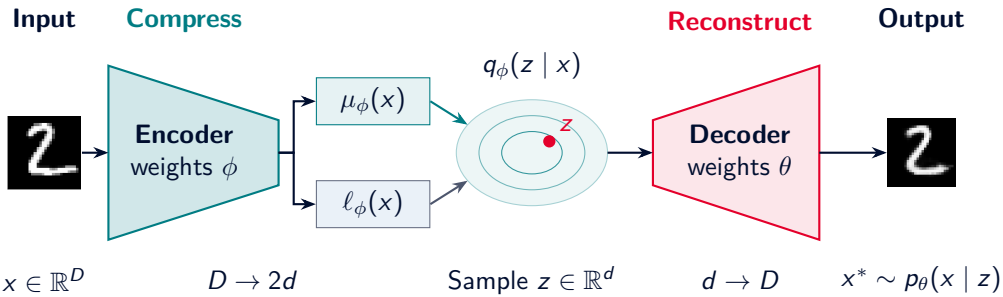
$$q_\phi(z \mid x) = \mathcal{N}\Big(\mu_\phi(x), \operatorname{diag}\{\sigma^2_{\phi,1}(x), \ldots, \sigma^2_{\phi,d}(x)\}\Big).$$



- **Amortized VI:** one network predicts a distribution for every input.
- The encoder outputs $2d$ numbers. A sampled latent vector still has dimension d.

Kingma & Welling (2014), Auto-Encoding Variational Bayes. $\exp(\ell_\phi)$ is applied elementwise.

VAE architecture



$$q_\phi(z \mid x) = \mathcal{N}(z; \mu_\phi(x), \text{diag}\{\exp(\ell_\phi(x))\}),$$
$$p_\theta(x \mid z) = \mathcal{N}\Big(x; f_\theta(z), \tau^2 I_D\Big).$$

VI recap I

Fix x and θ. Choose the variational approximation by minimizing

$$\phi^\star = \arg\min_\phi \mathrm{KL}\{q_\phi(z \mid x) \,\|\, p_\theta(z \mid x)\}\,.$$

Using

$$\log p_\theta(z \mid x) = \log p_\theta(x, z) - \log p_\theta(x),$$

we obtain

$$\begin{aligned}\mathrm{KL}\{q_\phi \| p_\theta(\cdot \mid x)\} &= \mathbb{E}_{q_\phi}[\log q_\phi(Z \mid x) - \log p_\theta(x, Z)] + \log p_\theta(x) \\ &= \log p_\theta(x) - \mathcal{L}_x(\theta, \phi),\end{aligned}$$

where the evidence lower bound (ELBO) is

$$\mathcal{L}_x(\theta, \phi) = \mathbb{E}_{q_\phi}[\log p_\theta(x, Z) - \log q_\phi(Z \mid x)]\,.$$

Since $\log p_\theta(x)$ does not depend on ϕ,

$$\boxed{\arg\min_\phi \mathrm{KL}\{q_\phi \| p_\theta(\cdot \mid x)\} = \arg\max_\phi \mathcal{L}_x(\theta, \phi).}$$

VI recap II

Using the joint-model factorization

$$p_\theta(x, z) = p_\theta(x \mid z)p(z),$$

the ELBO can also be written as

$$\begin{aligned}\mathcal{L}_x(\theta, \phi) &= \mathbb{E}_{q_\phi}[\log p_\theta(x \mid Z) + \log p(Z) - \log q_\phi(Z \mid x)] \\ &= \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(x \mid Z)]}_{\text{expected log likelihood}} - \underbrace{\mathrm{KL}\{q_\phi(z \mid x) \,\|\, p(z)\}}_{\text{prior KL}}.\end{aligned}$$

- The expected log likelihood rewards latent draws that explain the observed x.
- The prior KL regularizes $q_\phi(z \mid x)$ toward the latent prior $p(z)$.

Closed-form Gaussian prior KL

For a fixed image x, write $\mu = \mu_\phi(x)$ and $\sigma^2 = \sigma^2_\phi(x) = \exp(\ell_\phi(x))$:

$$q_\phi(z \mid x) = \mathcal{N}\Big(z; \mu, \mathrm{diag}(\sigma^2)\Big), \qquad p(z) = \mathcal{N}(z; 0, I_d).$$

The log-density ratio is

$$\log \frac{q_\phi(z \mid x)}{p(z)} = \frac{1}{2} \sum_{j=1}^{d} \left[-\log \sigma_j^2 - \frac{(z_j - \mu_j)^2}{\sigma_j^2} + z_j^2 \right].$$

Under $q_\phi(z \mid x)$,

$$\mathbb{E}_{q_\phi}[(Z_j - \mu_j)^2] = \sigma_j^2, \qquad \mathbb{E}_{q_\phi}[Z_j^2] = \mu_j^2 + \sigma_j^2.$$

Taking expectations gives the **prior KL inside the ELBO**:

$$\boxed{\mathrm{KL}\{q_\phi(z \mid x) \,\|\, p(z)\} = \frac{1}{2} \sum_{j=1}^{d} \left(\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2 \right).}$$

VAE training

The Gaussian prior KL has a closed form, but the expected log likelihood is generally intractable for a nonlinear decoder. By **Monte Carlo** approximation:

$$\mathbb{E}_{q_\phi(z|x_i)}[\log p_\theta(x_i \mid Z)] \approx \frac{1}{S}\sum_{s=1}^{S} \log p_\theta(x_i \mid z_i^{(s)}),$$

where $z_i^{(1)}, \ldots, z_i^{(S)} \overset{\text{i.i.d.}}{\sim} q_\phi(z \mid x_i)$.

Minibatch objective: For a minibatch $\mathcal{B}$ of size B, the estimated negative ELBO is

$$\widehat{\mathcal{J}}_{\mathcal{B}}(\theta, \phi) = \frac{1}{B}\sum_{i \in \mathcal{B}} \left[-\frac{1}{S}\sum_{s=1}^{S} \log p_\theta(x_i \mid z_i^{(s)}) + \mathrm{KL}\{q_\phi(z \mid x_i) \,\|\, p(z)\} \right].$$

In practice, $S = 1$ is often used: **one latent draw per observation in each minibatch**.

Gradients through sampling

To train the encoder, we need the gradient of the **expected log likelihood**:

$$\nabla_\phi \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid Z)] \,.$$

Here ϕ affects the expectation through the **distribution of** Z.

What goes wrong with naive sampling?
Suppose we draw $z^{(s)} \sim q_\phi(z \mid x)$ and treat these sampled values as constants during backpropagation. Then

$$\nabla_\phi \left[\frac{1}{S} \sum_{s=1}^{S} \log p_\theta(x \mid z^{(s)}) \right] = 0, \qquad \text{with all } z^{(s)} \text{ held fixed.}$$

The decoder likelihood has no explicit dependence on ϕ, so this calculation misses how the encoder changes the latent distribution.

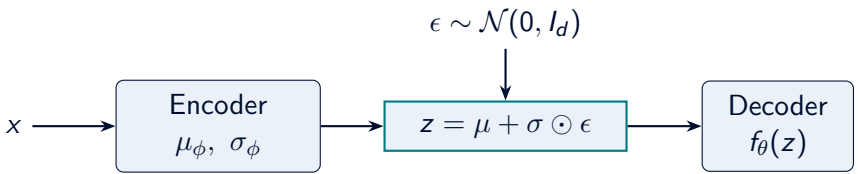
Reparameterization trick

To use ordinary backpropagation, we need a differentiable path from the encoder parameters to the latent draw.

Draw noise independent of the encoder weights:

$$\epsilon \sim \mathcal{N}(0, I_d), \qquad \boxed{Z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon = g_\phi(x, \epsilon).}$$

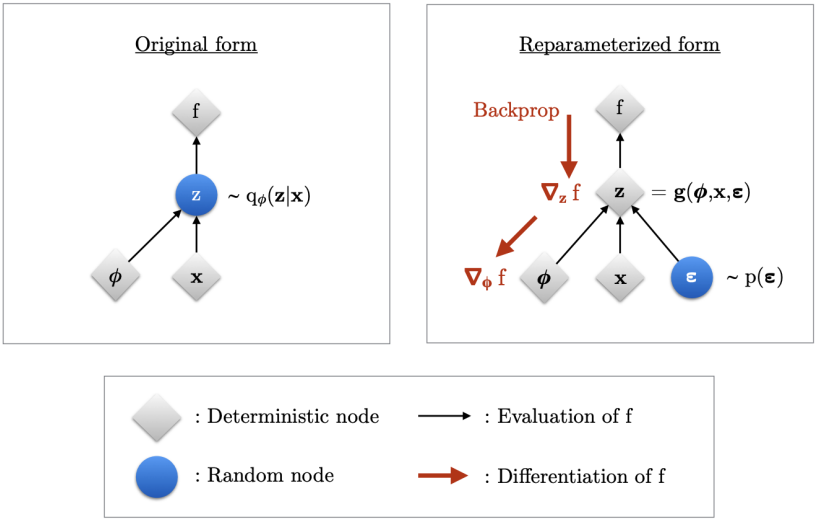
Then $Z \sim q_\phi(z \mid x)$ exactly. The product $\odot$ is elementwise.



For a fixed noise draw, z is differentiable in ϕ.

Kingma & Welling (2014), Auto-Encoding Variational Bayes.

Reparameterization trick



Reparameterization trick

Define $h_\theta(z) = \log p_\theta(x \mid z)$. Reparameterization moves the expectation to a fixed noise distribution:

$$\mathbb{E}_{q_\phi(z|x)}[h_\theta(Z)] = \mathbb{E}_{\epsilon\sim\mathcal{N}(0,I_d)}[h_\theta(g_\phi(x,\epsilon))]\,.$$

Because the noise distribution does not depend on ϕ, we can differentiate inside the expectation under standard regularity conditions:

$$\nabla_\phi \mathbb{E}_{q_\phi(z|x)}[h_\theta(Z)] = \mathbb{E}_{\epsilon\sim\mathcal{N}(0,I_d)}[\nabla_\phi h_\theta(g_\phi(x,\epsilon))]\,.$$

For independent draws $\epsilon^{(1)}, \ldots, \epsilon^{(S)} \sim \mathcal{N}(0, I_d)$, the Monte Carlo gradient estimator is

$$\boxed{\widehat{G}_\phi = \frac{1}{S}\sum_{s=1}^{S} \nabla_\phi h_\theta\left(g_\phi(x,\epsilon^{(s)})\right).}$$

It is **unbiased**, even for $S = 1$:

$$\mathbb{E}[\widehat{G}_\phi] = \nabla_\phi \mathbb{E}_{q_\phi(z|x)}[h_\theta(Z)].$$

Backpropagation computes each sample gradient with $\epsilon^{(s)}$ held fixed.

The reconstruction term

For the Gaussian decoder with fixed observation variance,

$$p_\theta(x \mid z) = \mathcal{N}(x; f_\theta(z), \tau^2 I_D).$$

Taking the negative log density gives

$$-\log p_\theta(x \mid z) = \frac{D}{2}\log(2\pi\tau^2) + \frac{1}{2\tau^2}\|x - f_\theta(z)\|^2.$$

Hence, up to a constant independent of θ and ϕ,

$$\boxed{-\mathbb{E}_{q_\phi}[\log p_\theta(x \mid Z)] = \frac{1}{2\tau^2}\mathbb{E}_\epsilon\|x - f_\theta(g_\phi(x, \epsilon))\|^2 + \text{const.}}$$

The reconstruction loss follows from the likelihood choice.

The VAE objective

For $S = 1$, draw $\epsilon_i \sim \mathcal{N}(0, I_d)$ and set

$$z_i = \mu_i + \exp(\ell_i/2) \odot \epsilon_i, \qquad (\mu_i, \ell_i) = (\mu_\phi(x_i), \ell_\phi(x_i)).$$

Minimize the Gaussian negative ELBO, dropping only the fixed constant:

$$\widehat{\mathcal{J}}_{\mathcal{B}}(\theta, \phi) = \frac{1}{B} \sum_{i \in \mathcal{B}} \left[\underbrace{\frac{\|x_i - f_\theta(z_i)\|^2}{2\tau^2}}_{\text{reconstruction term}} + \underbrace{\frac{1}{2} \sum_{j=1}^{d} \left(\mu_{ij}^2 + e^{\ell_{ij}} - 1 - \ell_{ij} \right)}_{\text{prior KL}} \right].$$

- Reconstruction encourages each latent draw to retain information about x_i.
- KL ties the inferred distributions to the prior used for generation.

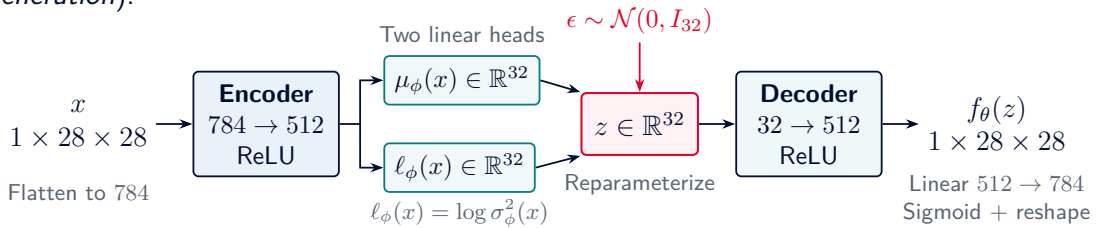
If reconstruction uses MSE per pixel, multiply it by $D/(2\tau^2)$ for this exact objective. An extra KL weight changes the relative weighting.

Try VAE on MNIST

Data: 28×28 grayscale images of handwritten digits, with pixels in $[0, 1]$.

Split: 55,000 training, 5,000 validation, and 10,000 test images.

Task: generate new digit images without specifying a digit label (*unconditional generation*).

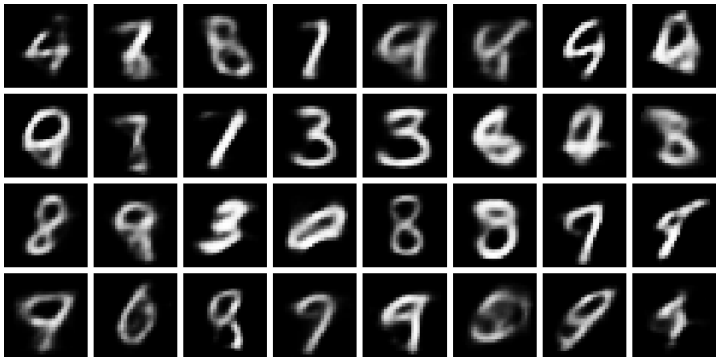


Training loss: squared error divided by $2\tau^2$, plus prior KL, with fixed $\tau^2 = 0.5$.

MNIST: generated digits

Draw from the prior and evaluate the decoder:

$$z \sim \mathcal{N}(0, I_{32}), \qquad \widetilde{x} = f_\theta(z) \in [0,1]^{1\times 28\times 28}.$$



New latent draws produce varied digit shapes; some digits are ambiguous.

Try VAE on CelebA

The CelebA data

- CelebA contains **202,599 face images**.
- We uses **50,000 training**, 5,000 validation, and 5,000 test images.
- Crop to 148×148; resize to 64×64 RGB; normalize to $[-1, 1]$.

The task: unconditional generation

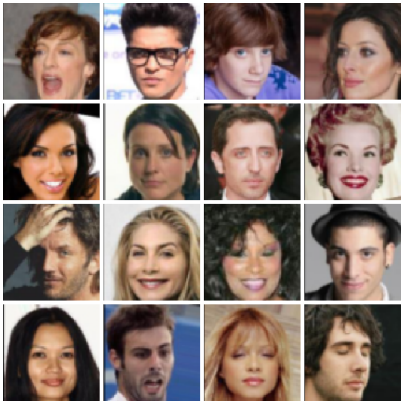
Learn the distribution of faces, then generate **new images from Gaussian latent vectors**.

$$x \in \mathbb{R}^{3\times 64\times 64}, \qquad D = 12{,}288.$$

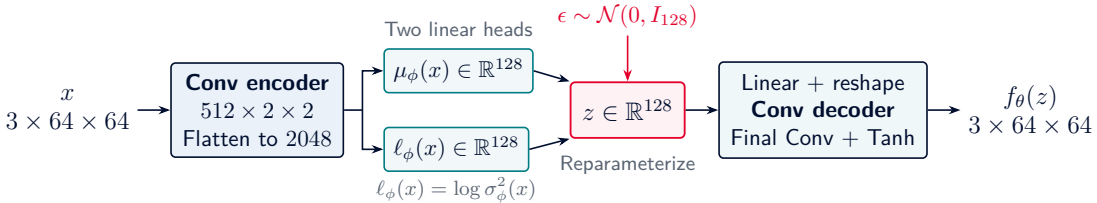
Attribute labels are unused.

Data: Liu et al., CelebA (2015).

Real held-out CelebA examples



Model structure for CelebA



Encoder

Five blocks: 3×3 convolution with stride 2, BatchNorm, and LeakyReLU.

$$C : 32 \to 64 \to 128 \to 256 \to 512,$$
$$H = W : 32 \to 16 \to 8 \to 4 \to 2.$$

Decoder

Linear $128 \to 2048$; reshape to $512 \times 2 \times 2$.
Five stride-2 transposed convolutions, each with BatchNorm and LeakyReLU.

$$C : 512 \to 256 \to 128 \to 64 \to 32,$$
$$H = W : 2 \to 4 \to 8 \to 16 \to 32.$$

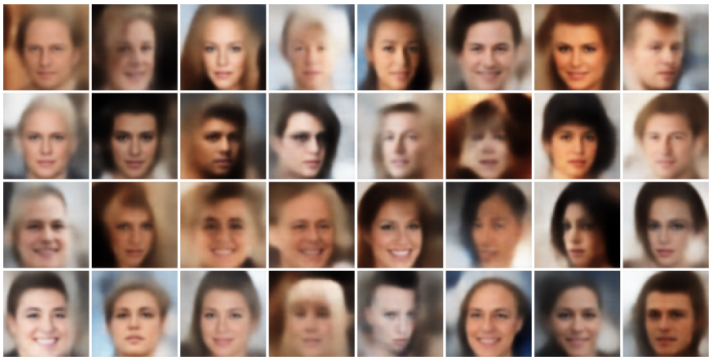
Final 3×3 Conv + Tanh: $3 \times 64 \times 64$.

Structure adapted from AntixK/PyTorch-VAE.

CelebA: generated faces

Draw from the prior and evaluate the decoder:

$$z \sim \mathcal{N}(0, I_{128}), \qquad \widetilde{x} = f_\theta(z) \in [-1, 1]^{3\times 64\times 64}.$$



Recognizable faces with **smooth textures and limited fine detail**.

VAEs produce blurry images

Fix the encoder $q_\phi(z \mid x)$, and let $p_{\text{data}}(x)$ denote the data distribution. Define

$$q_\phi(z) = \int p_{\text{data}}(x) q_\phi(z \mid x)\, dx, \qquad q_\phi(x \mid z) = \frac{p_{\text{data}}(x) q_\phi(z \mid x)}{q_\phi(z)}.$$

For a Gaussian decoder with fixed variance, minimize the squared error:

$$\mathcal{R}(f) = \mathbb{E}_{p_{\text{data}}(x)} \mathbb{E}_{q_\phi(z|x)} \left[\|x - f(z)\|^2 \right] = \mathbb{E}_{q_\phi(z)} \mathbb{E}_{q_\phi(x|z)} \left[\|x - f(z)\|^2 \right].$$

For each z, squared error is minimized by the conditional mean:

$$\boxed{f^*(z) = \arg\min_a \mathbb{E}_{q_\phi(x|z)} \left[\|X - a\|^2 \right] = \mathbb{E}_{q_\phi(x|z)}[X].}$$

Averaging images with different edges or textures can produce blur.

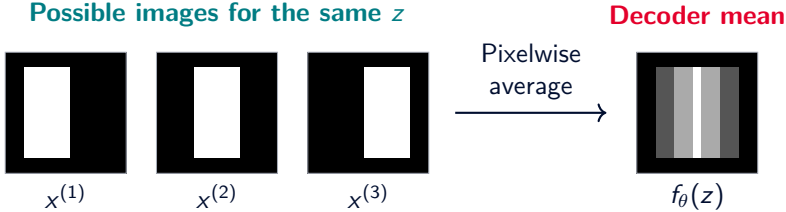
VAEs produce blurry images

For a fixed z, suppose $q_\phi(x \mid z)$ assigns equal probability to three images with slightly different stroke positions:

$$\Pr_{q_\phi}\Big(X = x^{(k)} \mid Z = z\Big) = \frac{1}{3}, \qquad k = 1, 2, 3.$$

Under squared error, the optimal decoder outputs their **pixelwise average**:

$$\widehat{x} = f^*(z) = \mathbb{E}_{q_\phi}[X \mid Z = z] = \frac{x^{(1)} + x^{(2)} + x^{(3)}}{3}.$$



β-VAE

We can generalize the ordinary VAE by allowing a tuning parameter β:

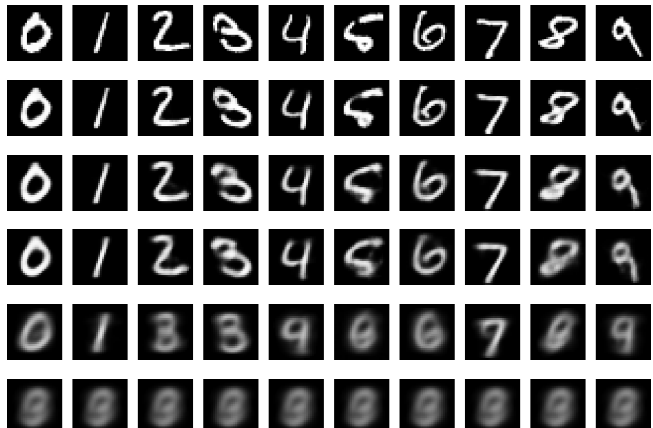
$$\mathcal{J}_\beta(\theta, \phi; x) = \underbrace{-\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid Z)]}_{\text{reconstruction}} + \beta \underbrace{\mathrm{KL}(q_\phi(z \mid x) \,\|\, p(z))}_{\text{regularization}}, \quad \beta \geq 0.$$

- $\beta = 0$: **reconstruction only**, use the same objective as an autoencoder. The latent codes are not constrained to match the prior.
- $\beta = 1$: the **ordinary VAE**.
- $\beta > 1$: stronger pressure toward the factorized prior can encourage **disentanglement**, often at the cost of reconstruction detail.

  Disentanglement means that different latent coordinates primarily control distinct factors of variation, such as stroke thickness or slant.
- $\beta \gg 1$: we have $q_\phi(z \mid x) \approx p(z)$, so the posterior carries little information about x.

β-VAE for reconstruction

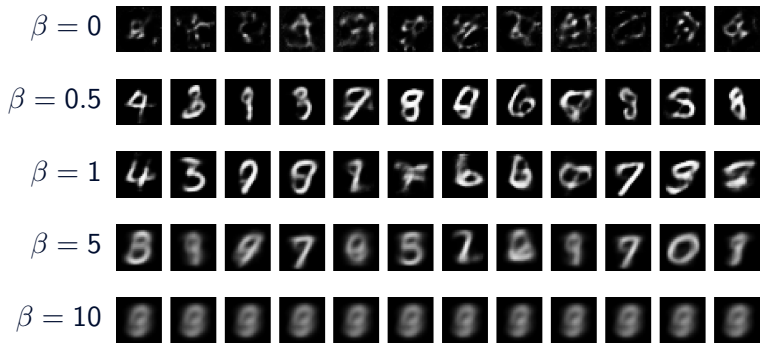
The same held-out digits; display $\hat{x} = f_\theta(\mu_\phi(x))$.



- $\beta = 0$: most detail.
- $\beta = 10$: nearly identical blurry outputs.
- Good reconstruction alone does not ensure good generation.

β-VAE for generation

Draw $z \sim \mathcal{N}(0, I_{32})$ and display the decoder mean $f_\theta(z)$.

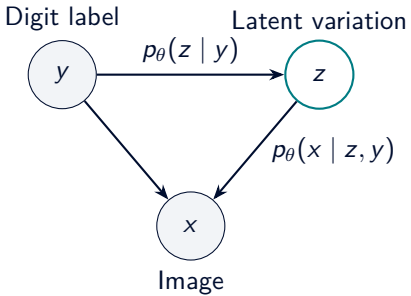


Columns reuse prior draws; digit labels need not match.

- $\beta = 0$: varied outputs, but poorly formed digits.
- $\beta = 10$: nearly identical, blurry outputs.
- Moderate β: more recognizable and varied digits.

Conditional VAE

Generate an image x for a requested digit label y. The latent variable z represents variation within the class.



Conditional generation

1. Draw $z \sim p_\theta(z \mid y)$.
2. Generate $x \sim p_\theta(x \mid z, y)$.

Training with image–label pairs

The encoder $q_\phi(z \mid x, y)$ infers z from the observed image and its label.

Sohn et al. (2015), Learning Structured Output Representation using Deep Conditional Generative Models.

CVAE: ELBO

Note that

$$p_\theta(x, z \mid y) = p_\theta(x \mid z, y) p_\theta(z \mid y).$$

Conditional evidence lower bound (ELBO)

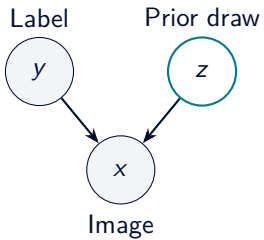
$$\begin{aligned}\mathcal{L}(\theta, \phi; x, y) &= \mathbb{E}_{q_\phi(z|x,y)}\left[\log \frac{p_\theta(x, z \mid y)}{q_\phi(z \mid x, y)}\right] \\ &= \mathbb{E}_{q_\phi(z|x,y)}[\log p_\theta(x \mid z, y)] - \mathrm{KL}(q_\phi(z \mid x, y) \,\|\, p_\theta(z \mid y)).\end{aligned}$$

KL divergence to the conditional prior

$$\mathrm{KL}(q_\phi(z \mid x, y) \,\|\, p_\theta(z \mid y)) = \mathbb{E}_{q_\phi(z|x,y)}\left[\log \frac{q_\phi(z \mid x, y)}{p_\theta(z \mid y)}\right] \geq 0.$$

CVAE with a fixed prior

Special case: $p_\theta(z \mid y) = p(z)$. Remove the $y \to z$ edge.



Prior: $y \perp\!\!\!\perp z$.

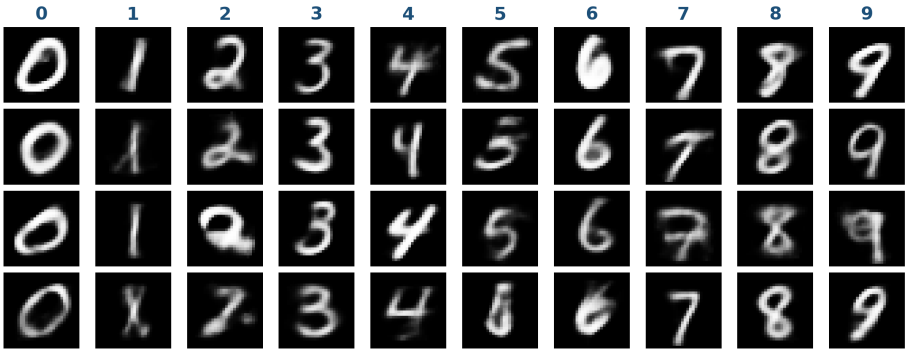
Conditional ELBO

$$\mathcal{L}(\theta, \phi; x, y) = \underbrace{\mathbb{E}_{q_\phi(z \mid x, y)}[\log p_\theta(x \mid Z, y)]}_{\text{conditional reconstruction}} - \underbrace{\mathrm{KL}(q_\phi(z \mid x, y) \,\|\, p(z))}_{\text{regularization toward the fixed prior}} .$$

The encoder still conditions on **both** x **and** y: prior independence does not imply posterior independence.

CVAE: MNIST generation

Choose $y \in \{0, 1, \ldots, 9\}$, draw $z \sim \mathcal{N}(0, I_{32})$, and display the decoder mean $f_\theta(z, y)$.



Columns: requested digit. Four independent latent draws per column.